

Generation Interconnection System Impact Study Report

For Queue Project Q16-0569

Valdosta 138 kV
December 22, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0569. The Interconnection Customer proposes a 287.0 MW wind facility interconnecting at the Valdosta 138 kV substation in the Pennsylvania area. The request was received on January 06, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 306 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$118,663,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Hot Wells-Roark 138kV line
Contingency: Flat Creek-Valdosta 138kV line
Loading: 156.2% (Rating: 174 MVA)
DFAX: 100.0%
Pre-project loading: 9.1%
2. Facility: Roark-Water St 138kV L
Contingency: Valdosta-Flat Crk 138kV
Loading: 137.1% (Rating: 202 MVA)
DFAX: 100.0%
Pre-project loading: 5.1%
3. Facility: Orchard Rd Substation 500 kV/345 kV Transformer
Contingency: Orchrdr Rd-F Humbug 500kV
Loading: 133.9% (Rating: 792 MVA)
DFAX: 5.2%
Pre-project loading: 132.0%
4. Facility: Water St-Valdosta 138kV
Contingency: Valdosta-Flat Crk 138kV
Loading: 129.5% (Rating: 218 MVA)
DFAX: 100.0%
Pre-project loading: 2.2%
5. Facility: Slate Creek-Hot Wells 138kV
Contingency: Valdosta-Flat Crk 138kV L
Loading: 128.9% (Rating: 204 MVA)
DFAX: 100.0%
Pre-project loading: 11.6%
6. Facility: Valdosta-Flat Creek 138kV
Contingency: Water St-Valdosta 138kV
Loading: 122.8% (Rating: 230 MVA)

- DFAX: 100.0%
Pre-project loading: 2.1%
7. Facility: Bair-Flat Crk 138kV
Contingency: Water St-Valdosta 138kV Line
Loading: 116.1% (Rating: 238 MVA)
DFAX: 100.0%
Pre-project loading: 4.6%
8. Facility: Blackrock-Shawville Sta 345kV
Contingency: Carbon Center 500-345 kV Auto
Loading: 108.3% (Rating: 1,694 MVA)
DFAX: 9.7%
Pre-project loading: 106.6%
9. Facility: Caldwell-Blackrock Substation 345kV
Contingency: Carbon Ctr-Caldwell 345kV
Loading: 105.7% (Rating: 996 MVA)
DFAX: 8.6%
Pre-project loading: 103.2%
10. Facility: Scott-Shawville Sta 345kV
Contingency: Orchard Rd-Shawville Station 500kV
Loading: 105.3% (Rating: 1,401 MVA)
DFAX: 6.7%
Pre-project loading: 104.0%
11. Facility: Caldwell-Carbon Ctr 345kV
Contingency: Blackrock-Shawville Sta 345kV
Loading: 103.7% (Rating: 1,258 MVA)
DFAX: 6.3%
Pre-project loading: 102.3%
12. Facility: Coronet-Ft Humbug 500kV Line
Contingency: Acadia-Montour Station 500kV
Loading: 103.6% (Rating: 2,819 MVA)
DFAX: 9.5%
Pre-project loading: 102.6%
13. Facility: Carbon Center 500/345kV Autotransformer
Contingency: Shawville Sta-Blackrock 345kV line
Loading: 103.1% (Rating: 1,447 MVA)
DFAX: 9.3%
Pre-project loading: 101.2%
14. Facility: Hilltop-Talenenergy Martins Creek Llc Williamsport 500kV
Contingency: Hilltop 500/345 kV Auto
Loading: 101.8% (Rating: 2,495 MVA)
DFAX: 10.7%
Pre-project loading: 100.5%
15. Facility: Strawberry Belt-Hilltop 345kV
Contingency: Hilltop-Talenenergy Martins Creek Llc Williamsport 500kV
Loading: 101.4% (Rating: 1,615 MVA)
DFAX: 6.2%
Pre-project loading: 100.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hot Wells-Roark 138kV line	Rebuild line with higher-capacity conductor	\$2,328,000	100% (\$2,328,000)
Roark-Water St 138kV L	Rebuild line with higher-capacity conductor	\$5,136,000	100% (\$5,136,000)
Orchard Rd Substation 500 kV/345 kV Transformer	Replace transformer with higher-rated unit	\$28,866,000	59% (\$16,970,000)
Water St-Valdosta Substation 138kV	Rebuild line with higher-capacity conductor	\$3,479,000	100% (\$3,479,000)
Slate Creek-Hot Wells 138kV	Rebuild line with higher-capacity conductor	\$5,257,000	100% (\$5,257,000)
Valdosta-Flat Creek 138kV	Rebuild line with higher-capacity conductor	\$4,645,000	100% (\$4,645,000)
Bair-Flat Crk 138kV	Rebuild line with higher-capacity conductor	\$19,858,000	100% (\$19,858,000)
Blackrock-Shawville Sta 345kV	Reconductor line	\$19,881,000	24% (\$4,768,000)
Caldwell-Blackrock Substation 345kV	Reconductor line	\$24,270,000	35% (\$8,436,000)
Shawville Sta-Scott 345kV L	Reconductor line	\$32,982,000	26% (\$8,661,000)
Carbon Center-Caldwell Substation 345kV line	Reconductor line	\$25,074,000	25% (\$6,162,000)
Coronet-Ft Humbug 500kV Line	Reconductor line	\$62,859,000	21% (\$13,181,000)
Carbon Center 500/345kV Autotransformer	Replace transformer with higher-rated unit	\$26,336,000	28% (\$7,494,000)
Hilltop-Talenenergy Martins Creek Llc Williamsport 500kV	Reconductor line	\$33,911,000	26% (\$8,869,000)
Strawberry Belt-Hilltop 345kV	Reconductor line	\$12,447,000	28% (\$3,419,000)
	Total allocated to project		\$118,663,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.